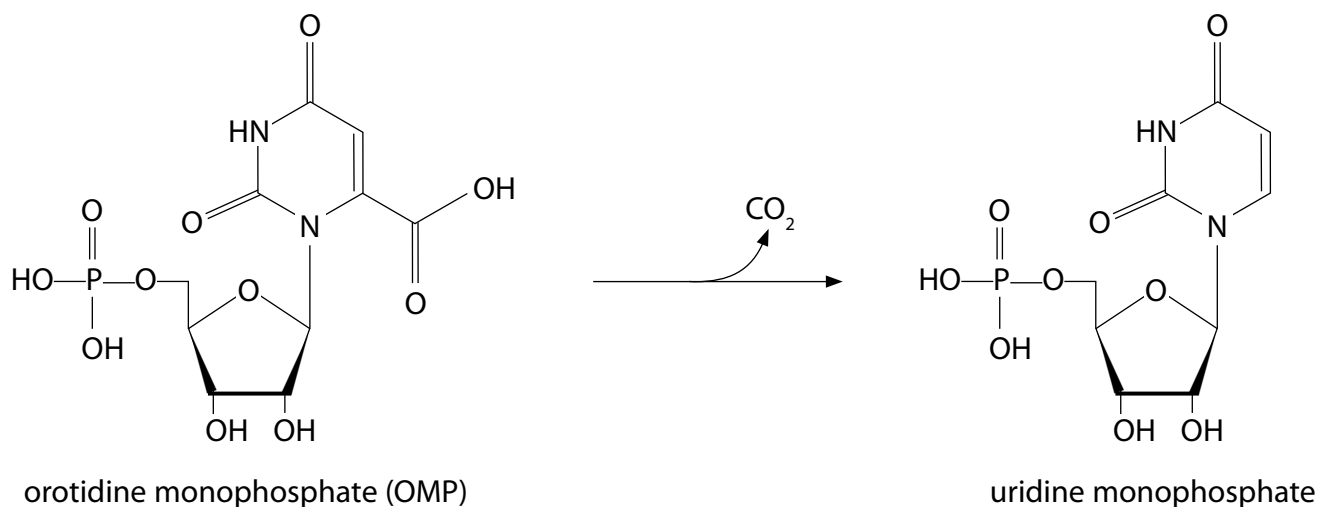


2 The enzyme OMP decarboxylase is involved in the synthesis of the mononucleotide uridine monophosphate.

(a) The diagram shows the reaction catalysed by OMP decarboxylase.



(i) Which sugar is present in uridine monophosphate?

(1)

- ☐ A deoxyribose
- ☐ B galactose
- ☐ C glucose
- ☐ D ribose

(ii) Draw a circle around the base in uridine monophosphate.

(1)

(iii) Suggest one way in which cells use uridine monophosphate.

(1)

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(b) The enzyme OMP decarboxylase increases the rate of carbon dioxide removal from orotidine monophosphate by 10^{17} times.

(i) State how OMP decarboxylase increases the rate of this reaction.

(2)

(ii) Explain why OMP decarboxylase catalyses this reaction only.

(3)

(Total for Question 2 = 8 marks)